

Test plan

This document sets out how PrintXpress is tested. It covers the approach, the environment, the test data, and the test cases. It maps to Task E of the assignment. Each test case links back to a requirement id from [03-requirements.md](#) so coverage is easy to check.

Approach

Testing happens at three levels, from small to large.

1. Unit tests check single pieces of logic in isolation, for example password hashing, price calculation, and promotion validity. These run on the JVM with JUnit and are fast.
2. Instrumented and UI tests check the app on a device or emulator, for example that a Room insert and read work, and that a screen shows the right content. These use AndroidX Test and Compose UI testing.
3. Manual tests follow the test cases below by hand on an emulator. They confirm the full user journeys end to end and catch anything the automated tests miss.

The aim is to cover every Must requirement with at least one test, and to test both the happy path and the common error paths such as a duplicate email or a missing file.

Environment

Item	Value
Device	Android emulator, Pixel 6, and one physical phone if available
Android version	API 24 (minimum) and a recent API for the main runs
Build	Debug build from Android Studio
Data	A fresh install with the seeded catalog, plus a test account created during testing
Tools	JUnit, AndroidX Test, Compose UI test, and manual checks

Test data

The same data sets are reused across cases so results are repeatable.

Set	Field	Value
Valid user	name, email, phone, password	Nimal Perera, nimal@example.com, 0771234567, Print123
Duplicate user	email	nimal@example.com (already registered)
Invalid email	email	nimal.example.com (no at sign)

Short password	password	123 (under six characters)
Valid address	label, line1, city, postal	Home, 25 Galle Road, Colombo, 00300
Valid order	product, quantity, size, material	Business cards, 100, A6, glossy
Design file	file	sample-card.png (a small valid image)
Promotion (active)	code, dates	FESTIVE10, valid now
Promotion (expired)	code, dates	OLD5, valid in the past

Test cases

Status starts as Not run and is set to Pass or Fail when the case is carried out.

Authentication and profile

Id	Requirement	Steps	Test data	Expected result	Status
TC-01	FR-01, FR-03	Register a new account	Valid user	Account is created, password is stored hashed, home screen opens	Not run

TC-02	FR-02	Register with an email already in use	Duplicate user	Register is blocked with a clear message	Not run
TC-03	FR-01, NFR-08	Register with a bad email	Invalid email	Form shows a validation error and does not save	Not run
TC-04	NFR-08	Register with a short password	Short password	Form shows a validation error and does not save	Not run
TC-05	FR-04	Log in with correct details	Valid user	Login succeeds and home screen opens	Not run
TC-06	FR-04	Log in with a wrong password	Valid user, wrong password	Login fails with one clear message	Not run
TC-07	FR-05	Close and reopen the app while logged in	Valid user	The app opens still logged in	Not run
TC-08	FR-06	Edit the profile name and save	Valid user	The new name shows and is saved	Not run
TC-09	FR-07	Add an address and set it as default	Valid address	The address is saved and marked default	Not run
TC-10	FR-08	Log out	Valid user	Login state clears and the login screen shows	Not run

Id	Requirement	Steps	Test data	Expected result	Status
TC-11	FR-09	Open the home screen	Seeded catalog	Product categories show, loaded from the local database	Not run
TC-12	FR-10	Open a category	Business cards	The product list for that category shows	Not run
TC-13	FR-11	Open a product	Business cards	Name, price, material, sample image, and options show	Not run
TC-14	FR-12	Filter products by material	glossy	The list updates to matching products, and clearing it restores the full list	Not run

Ordering

Id	Requirement	Steps	Test data	Expected result	Status
TC-15	FR-13	Upload a design file	Design file	The chosen file name shows and is saved with the item	Not run
TC-16	FR-14	Enter custom text instead of a file	Text only	The text is accepted and saved with the item	Not run

TC-17	FR-15	Try to add an item with no file and no text	Empty	The app blocks it with a clear message	Not run
TC-18	FR-16	Change quantity and material	Valid order	The price updates to match the selection	Not run
TC-19	FR-17, FR-18	Place a delivery order with an address	Valid order, valid address	The order saves with status processing and a delivery address	Not run
TC-20	FR-19	View the confirmation after placing an order	Valid order	The confirmation shows the order number and a summary	Not run
TC-21	FR-17	Try to place a delivery order with no address	Valid order, no address	The app blocks it and asks for an address	Not run
TC-22	FR-20	Save a design and reuse it on a new order	Design file	The saved design appears and can be picked again	Not run

Order management

Id	Requirement	Steps	Test data	Expected result	Status
TC-23	FR-21, FR-22	Open the orders screen	Existing orders	Orders show with date, total, and status	Not run

TC-24	FR-23	Cancel an order that is still processing	Processing order	The order moves to cancelled	Not run
TC-25	FR-24	Try to cancel an order that is printing	Printing order	Cancel is hidden or disabled with a short note	Not run
TC-26	FR-23	Reschedule an order that is still processing	Processing order	The reschedule is accepted and saved	Not run

Notifications and offers

Id	Requirement	Steps	Test data	Expected result	Status
TC-27	FR-25, FR-26	Place an order and open notifications	Valid order	A confirmation notification shows as unread, and opening it marks it read	Not run
TC-28	FR-27	Open the offers screen	Active promotion	The active offer shows with title, discount, code, and dates	Not run
TC-29	FR-27	Check an expired offer	Expired promotion	The expired offer does not show	Not run

Help and support

Id	Requirement	Steps	Test data	Expected result	Status
TC-30	FR-28	Open print guidelines	None	Guidance on file types, resolution, colour mode, and bleed shows	Not run
TC-31	FR-29	Open FAQs and contact	None	The FAQ list and the shop contact details show	Not run

Non-functional checks

Id	Requirement	Steps	Expected result	Status
TC-32	NFR-02	Turn off the network and use the app	Every screen works with no network	Not run
TC-33	NFR-07	Force an error such as a duplicate email	The app shows a message and does not crash	Not run
TC-34	NFR-05	Check touch targets and content descriptions	Buttons are easy to tap and icons have descriptions	Not run
TC-35	NFR-10	Export the database file and open it in DB Browser for SQLite	The tables and rows are visible	Not run

Defect handling

When a case fails, record the test id, what happened, the steps to reproduce, and a screenshot. Fix the cause, then run the failed case again and any related cases before marking it Pass.

Exit criteria

Testing is complete when every Must requirement has at least one passing test, no known defect remains in a Must feature, and the offline, error handling, and database export checks all pass.